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Central Bank Liquidity Reallocation and Bank Lending: Evidence from the Tiering System

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1 Big picture

- **Question.** Do central bank reserves stimulate bank lending, and does the answer depend not just on the *quantity* of reserves in the banking system but on *which banks* hold them?
- **Setting.** The paper studies euro-area banks around the ECB’s tiered reserve remuneration reform. The key policy decision was announced on September 12, 2019 and implemented on October 30, 2019, while the first hint that tiering was coming arrived in Mario Draghi’s March 27, 2019 speech.
- **Policy shock.** The ECB exempted excess liquidity holdings up to six times a bank’s minimum reserve requirement from the negative deposit facility rate. That change raised the marginal return on reserves only for banks whose reserve holdings were initially below their exemption allowance.
- **Core empirical object.** The main treatment is a bank’s unused exemption allowance,

$$\text{Exposure}_i = \max \left\{ \frac{6\text{MRR}_i - \text{EL}_i}{\text{Assets}_i}, 0 \right\},$$

measured just before the hint of tiering in February 2019 and just before implementation in October 2019.

- **Main reserve-allocation result.** Banks with larger unused allowances accumulated more excess liquidity after implementation. A one-percentage-point larger unused allowance is associated with a monthly increase in excess liquidity of about 0.224 percentage points of assets in the post-implementation period.
- **How reserves moved.** The reallocation occurred mainly through the money market. Banks with unused allowances borrowed more in both secured and unsecured markets, while banks already flush with reserves lent more without charging meaningfully higher rates.
- **Main lending result.** Once the authors control for credit demand using high-dimensional borrower-time fixed effects, banks with larger unused allowances extend more credit, start more new lending relationships, cut lending rates if they were funding-constrained ex ante, lengthen maturities, and expand credit lines.
- **Balance-sheet mechanism.** The banks that obtained reserves also reduced holdings of government securities. The idea is that reserves provide superior liquidity insurance, so banks need to hoard fewer liquid securities and can reallocate balance-sheet capacity toward loans.
- **Who drives the effect?** The lending response is concentrated among banks that looked ex ante financially constrained: high money-market borrowing rates, low capitalization, or high CDS spreads. That is exactly what the paper’s conceptual framework predicts.
- **Aggregate conclusion.** Reallocating one euro of reserves toward banks with initially low reserve holdings increases aggregate credit supply by roughly 15 cents.
- **Why the paper matters.** The paper argues that reserve *distribution* matters for monetary transmission even when the aggregate stock of reserves is abundant. That is directly relevant for quantitative tightening: draining reserves from already liquidity-constrained banks can be much more contractionary than draining reserves from satiated banks.

2 Data and measurement (key objects)

2.1 Bank-level data and policy variables

- **IBSI balance-sheet data.** The Individual Balance Sheet Indicators dataset gives monthly balance-sheet information for 128 euro-area banks from January 2014 to February 2020. It contains loans, deposits, securities, and other balance-sheet items.
- **ECB liquidity data.** The authors observe banks’ take-up of targeted longer-term refinancing operations (TLTRO s) from the ECB’s proprietary liquidity records.
- **Market-based funding-cost measures.** Bank CDS spreads and stock prices come from Thomson Reuters. CDS spreads are crucial because the conceptual framework says the treatment should matter most for high-cost-of-capital banks.
- **Main bank-level outcome variables.** Monthly changes in excess liquidity, government-securities holdings, and outstanding loans to nonfinancial corporations.
- **Excess liquidity.** Excess liquidity is defined as current account holdings plus the deposit facility minus minimum reserve requirements.
- **Treatment intensity.** The October 2019 exposure variable is the unused exemption allowance right before implementation. The February 2019 exposure variable is the corresponding “notional” treatment intensity before tiering was formally designed, used to check for anticipation effects.
- **Timing dummies.** Interim_t equals one between the March 2019 hint and the October 2019 implementation. Implementation_t equals one after October 30, 2019, when the tiering system actually took effect.

2.2 Loan-level and firm-level data

- **AnaCredit.** The core lending dataset is AnaCredit, a harmonized euro-area credit register. The sample used for credit-supply tests has 122 banks, 2,624,856 firms, and 3,439,580 bank-firm relationships from September 2018 to February 2020.
- **Geographic and industry detail.** Borrowers span 19 euro-area countries, 89 two-digit NACE industries, and 1,055 NUTS2 regions. Firms are also assigned to borrower-size deciles.
- **Why this matters.** The granularity lets the authors compare how two banks lend to the same firm, or at least to firms in the same industry-location-size-time cell, so they can separate credit supply from credit demand.
- **Loan outcomes.** Outstanding loan volume, new relationships, terminated relationships, lending rates, loan maturity, and drawn/undrawn credit lines.

2.3 Money-market data

- **MMSR.** The Money Market Statistical Reporting dataset contains daily transaction-level information on euro-area money-market trades. Around 50 large banks report all secured and unsecured transactions up to 12 months’ maturity.

- **Scale.** During the sample period, the raw data contain about 30 million secured transactions and 12 million unsecured transactions. The estimating sample used for daily bank-level regressions has 42 banks and 44,269 bank-day observations.
- **Main outcomes.** Stocks of gross borrowing, gross lending, and net borrowing in secured and unsecured markets, plus borrowing rates and maturities.
- **Normalization.** Money-market positions are scaled by a bank’s minimum reserve requirement so treatment effects can be interpreted in “units” of the tiering allowance.

2.4 Auxiliary measures

- **Tiering benefits.** To test a pure profitability channel, the paper constructs the hypothetical savings created by the tiering rule for each bank,

$$\text{TieringBenefits}_i = \frac{\min\{0, \text{DFR}(\text{EL}_i - 6\text{MRR}_i)\} - \text{DFREL}_i}{\text{Assets}_i}.$$

Banks with large reserve holdings but no unused allowances could still enjoy accounting savings from tiering; if the results were driven by profitability rather than reserve reallocation, these banks should also expand lending.

- **Funding-cost heterogeneity.** The paper splits banks by pre-implementation secured money-market borrowing rates, CDS spreads, and capitalization to test whether the effect is strongest for banks that previously found reserves expensive to hold.
- **Borrower heterogeneity.** The authors also split firms by default probability, size, profitability, leverage, and productivity to test whether additional credit went disproportionately to risky or inefficient borrowers.

3 Models and empirical specifications (all core equations + interpretation)

3.1 Conceptual framework: reserves, securities, and lending

The paper models a bank that allocates its balance sheet across reserves (R), securities (S), and loans (L). A simple reduced-form profit function is

$$\pi = r(R)R + r(S)S + r(L)L + v(R, L) + u(R, S) - c(R + S + L), \quad (1)$$

where $v(R, L)$ captures the liquidity-insurance value of reserves for lending, and $u(R, S)$ captures the idea that securities are second-best substitutes for reserves.

$$\begin{aligned} r(R) + v_R + u_R - c &= 0, \\ r(L) + v_L - c &= 0, \\ r(S) + u_S - c &= 0. \end{aligned} \quad (2)$$

- **Interpretation.** Reserves make lending safer because they insure banks against liquidity shocks. The cross-partial $v_{RL} > 0$ means that when a bank holds more reserves, loans become more attractive at the margin.

- **Satiation point.** The paper assumes a reserve satiation point above which the marginal convenience yield of reserves becomes essentially flat. Banks already above that point are willing to lend reserves to other banks without demanding meaningfully higher rates.
- **Key comparative statics.** A bank with high funding costs optimally holds too few reserves and therefore too few loans. Raising the marginal return on reserves for that bank increases reserve demand, lowers desired securities holdings, and increases lending.
- **Economic punchline.** The treatment should not matter uniformly across banks. It should matter most for low-reserve, high-funding-cost banks.

3.2 Exposure construction and treatment intensity

The treatment variable is the bank's unused exemption allowance relative to assets:

$$\text{Exposure}_i^\tau = \max \left\{ \frac{6\text{MRR}_i^\tau - \text{EL}_i^\tau}{\text{Assets}_i^\tau}, 0 \right\}, \quad \tau \in \{\text{Feb 2019}, \text{Oct 2019}\}. \quad (3)$$

- **Interpretation.** Banks with low reserve holdings relative to their exempt tier have positive exposure: they stand to gain from acquiring more reserves once the tiering system is active.
- **Why this is clever.** The ECB did not allocate exemptions based on banks' current reserve positions. Instead, the exemption is a multiple of minimum reserve requirements, which are predetermined by deposit-related rules. That creates plausibly exogenous cross-bank variation in the increase in the marginal return on reserves.

3.3 Excess-liquidity regression

The first step is a difference-in-differences regression for reserve accumulation:

$$\Delta \frac{\text{EL}_{ict}}{\text{Assets}_{ict}} = \beta_1 (\text{Interim}_t \times \text{Exposure}_i^{\text{Feb 2019}}) + \beta_2 (\text{Implementation}_t \times \text{Exposure}_i^{\text{Oct 2019}}) + \beta_3 \text{CDS}_{it} + \alpha_i + \alpha_{ct} + u_{ict}. \quad (4)$$

- **Dependent variable.** Monthly change in excess liquidity relative to assets.
- **Interpretation of β_2 .** If the tiering system really raises reserve demand only for banks with unused allowances, β_2 should be positive and larger than β_1 .
- **What the paper finds.** That is exactly what happens: the post-implementation coefficient is about three times the interim coefficient.

3.4 Money-market reallocation regression

To show how reserves move across banks, the paper estimates daily money-market regressions of the form

$$\text{MoneyMarketActivity}_{ictm} = \beta_1 (\text{Interim}_t \times \text{Exposure}_i^{\text{Feb 2019}}) + \beta_2 (\text{Implementation}_t \times \text{Exposure}_i^{\text{Oct 2019}}) + \beta_3 \text{CDS}_{it} + \alpha_i + \alpha_{cm} + u_{ictm}. \quad (5)$$

- **Dependent variable.** One of six daily outcomes: gross borrowing, gross lending, or net borrowing in secured or unsecured money markets.
- **Scaling.** Outcomes are expressed relative to minimum reserve requirements, which makes the coefficients directly comparable to the tiering allowance.
- **Interpretation.** A positive β_2 on net borrowing means the treated banks fill their unused allowances by borrowing reserves from other banks rather than from the ECB.
- **Auxiliary implication.** If the supply of reserves from high-liquidity banks is elastic, borrowing rates and maturities should not deteriorate much for the treated banks.

3.5 Loan-supply regression with credit-demand controls

The main lending specification is

$$\text{Loan}_{f,i,t} = \beta_1(\text{Interim}_t \times \text{Exposure}_i^{\text{Feb 2019}}) + \beta_2(\text{Implementation}_t \times \text{Exposure}_i^{\text{Oct 2019}}) + \beta_3 X_{i,t} + \gamma_{f,t} + \delta_{i,f} + \varepsilon_{f,i,t}, \quad (6)$$

where $\text{Loan}_{f,i,t}$ is the outstanding credit from bank i to firm f , $X_{i,t}$ contains bank controls, $\gamma_{f,t}$ are firm-time fixed effects (or industry-location-size-time fixed effects), and $\delta_{i,f}$ are bank-firm fixed effects in the strongest specification.

- **Interpretation of β_2 .** This is the causal lending-supply effect of an exogenous increase in the marginal value of reserves, net of borrower demand.
- **Why firm-time fixed effects matter.** They compare how two banks change lending to the same borrower in the same month. That is the cleanest way to purge borrower-level demand shocks.
- **Bank controls.** Excess liquidity, government securities, deposit ratio, TLTRO funds, and CDS spreads absorb other channels linked to balance-sheet strength and concurrent monetary policies.

3.6 Alternative-channel and firm-level aggregation equations

The paper also uses two additional constructions.

First, the profitability channel is tested with TieringBenefits_i , defined above. If pure accounting relief drives lending, $\text{TieringBenefits}_i \times \text{Implementation}_t$ should matter even for banks that do not acquire reserves.

Second, firm-level borrowing is studied with firm-specific averages of bank exposures:

$$\overline{\text{Exposure}}_{f,t} = \sum_i \omega_{f,i,t-1} \text{Exposure}_i, \quad (7)$$

where $\omega_{f,i,t-1}$ is the lagged loan share of bank i in firm f 's borrowing.

- **Interpretation.** This aggregation asks whether firms connected to more strongly treated banks end up borrowing more in total, not just within a given bank-firm pair.
- **Why this matters.** If reserve reallocation simply reshuffled lending across banks without increasing aggregate credit, the firm-level effect should be small. The paper finds the opposite.

4 Identification and credibility

4.1 Why the tiering reform is a quasi-natural experiment

- **Exemptions were based on MRR, not on ex ante reserve hoards.** The ECB exempted up to six times minimum reserve requirements from the negative deposit facility rate. Since MRR is tied to deposit-related rules, it is predetermined relative to the tiering announcement.
- **Only some banks got a higher marginal return on reserves.** Banks already holding enough reserves to fill the exempt tier saw no change in the marginal return to an extra euro of reserves. Banks with unused allowances did.
- **This isolates reserve demand.** The treatment shifts the demand for reserves for reasons unrelated to contemporaneous shocks to corporate loan demand.

4.2 Key comparison

- **Treated banks.** Banks with low reserve holdings relative to their exemption allowance.
- **Control banks.** Banks with high reserve holdings and little or no unused allowance.
- **Economic assumption.** Control banks are close to reserve satiation. Therefore they can transfer reserves through the money market without cutting loan supply or demanding sharply higher rates.

4.3 How the paper addresses major threats

- **Borrower demand shocks.** The authors use firm-month fixed effects or industry-location-size-month fixed effects, plus bank-firm fixed effects. That is the centerpiece of the loan-supply identification.
- **Concurrent ECB policies.** The September 2019 package also included a DFR cut, renewed asset purchases, and easier TLTRO terms. The regressions control for excess liquidity, deposit ratios, government-securities holdings, and TLTRO take-up; the results are robust to interacting these controls with the post period.
- **Profitability channel instead of reserve channel.** Banks with large ex ante reserve holdings benefit mechanically from tiering too, but those banks do *not* increase lending. The interaction with TieringBenefits is insignificant.
- **Pre-trends.** Interim-period coefficients are small, and the dynamic plot shows the effect turning positive only after implementation, not before.
- **Money-market crowding or rationing.** Treated banks do not face meaningfully higher borrowing rates or shorter maturities in money markets, which supports the reserve-satiation assumption on the lender side.

4.4 Limitation

- **What the design still needs.** The aggregate lending interpretation relies on the claim that high-reserve banks are satiated and therefore do not reduce lending when they transfer reserves.

- **Why the authors think this is reasonable.** They show no higher money-market rates, no large shifts in central-bank borrowing, and no evidence that high-reserve or high-tiering-benefit banks contract loan supply after implementation.

5 Tables and figures

5.1 Figure 1: Stock market reaction to the first mention of tiering

Read:

- EuroStoxx bank stocks jumped when Draghi first mentioned mitigating measures for negative-rate side effects, and they jumped again when Reuters reported that the ECB was working on a tiering system.
- The narrower bank index rose by almost 3% on the day, far more than the broader EuroStoxx 50.
- The figure shows that markets immediately understood tiering as a bank-relevant liquidity policy rather than a generic macro announcement.

5.2 Table 1: Summary statistics

Read:

- The bank-balance-sheet sample is large and long enough to study reserve adjustment: 128 banks over 74 months.
- The loan-level sample is exceptionally rich: over 52 million bank-firm-month observations after merging treatment variables and controls.
- The money-market sample confirms that the banks observed in MMSR are big enough players for the reserve-reallocation channel to show up clearly in market trading.

5.3 Figure 2 and Table 2: Excess-liquidity accumulation after tiering

Read:

- Figure 2 shows that after implementation the distribution of reserve holdings relative to exemptions compresses toward the 100% usage region. In plain language, banks filled their unused allowances.
- Table 2 quantifies this. The interim coefficient is 0.078, but the implementation coefficient is 0.224, both for a one-percentage-point larger unused allowance.
- The economic meaning is that the policy really changed reserve demand only when the reform became operative, not just when it was rumored.

5.4 Figures 3 and 4 and Table 3: The money market is the conduit

Read:

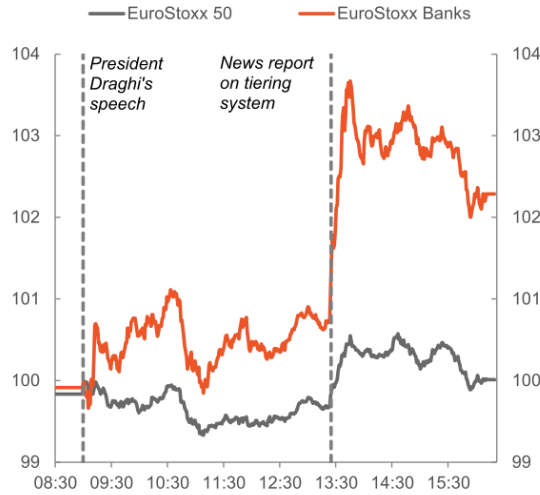


Fig. 1. Stock market reaction to the first mention of a tiering system
The chart shows the intraday development in the broad EuroStoxx50 index, as well as the narrow EuroStoxx Banks index on March 27, 2019, normalised to 100 at the start of trading at 9am. Former ECB president Draghi's speech containing a reference to "mitigating measures" to address the possible side effects of negative interest rates on bank profitability was released at 9:00 am in the morning and followed by an uptick in the EuroStoxx banks index of around 1 %, while the broader index remained largely unchanged. The release of a news bulletin reporting that the ECB was working on a tiering system at 13:25 was followed by an additional increase in banks' equity prices by around 2.5 %, compared to a rise of 0.7 % in the broader equity index.

Table 1

Summary statistics

Panel A summarizes the bank level dataset. We report observations at the bank and month level. Our sample consists of a panel of 128 banks from January 2014 to February 2020 (74 months). Panel B summarizes the Anacredit sample. We report observations at the bank, firm and month level. The Anacredit sample consists of a panel of 122 banks and 2,624,856 firms, for a total of 3,439,580 bank-firm relations, from September 2018 to February 2020 (18 months). Firms are distributed across 19 countries (AT, BE, CY, DE, EE, ES, FI, FR, GR, IE, IT, LT, LU, LV, MT, NL, PT, SI, SK), 89 2-digit NACE industries and 1,055 NUTS2 locations, providing 3,087,276 industry-location-size-month fixed effects. Panel C summarizes the MMSR sample.

Panel A. Bank-month level sample					
Variable name	Units	Definition	Obs.	Mean	St. Dev.
Monthly change in NFC loans	p.p.	Monthly change in ratio of NFC loans over assets.	9325	-0.004	0.878
Monthly change in excess liquidity	p.p.	Monthly change in ratio of excess liquidity (current account plus deposit facility minus minimum reserve requirements) over assets	9325	0.103	1.333
Monthly change in holdings of government securities	p.p.	Monthly change in ratio of holdings of government bonds over assets.	9325	-0.012	0.418
Exposure(Feb 2019)	%	Unused exemption allowance, i.e., difference of 6-fold the minimum reserve requirement and the excess liquidity holdings of a bank in February 2019 if such difference is positive, and to zero otherwise. It is expressed in percentage of main assets.	9325	0.879	1.480
Exposure(Oct 2019)	%	Unused exemption allowance, i.e., difference of 6-fold the minimum reserve requirement and the excess liquidity holdings of a bank in October 2019 if such difference is positive, and to zero otherwise. It is expressed in percentage of main assets.	9325	0.841	1.446
Interim(Mar-Oct 2019)	Cat.	Dummy variable equal to 1 if it is between March 2019 and October 2019, 0 otherwise.	9325	0.110	0.313
Implementation(Nov 2019-Feb 2020)	Cat.	Dummy variable equal to 1 if it is between November 2019 and February 2020, 0 otherwise.	9325	0.055	0.227
CDS	p.p.	5-years credit default swaps, in percentage points. One month lag.	9325	1.356	2.072
Panel B. Bank-firm-month level sample					
Variable name	Units	Definition	Obs.	Mean	St. Dev.
Volume of NFC loans	log(EUR mln)	Logarithm of outstanding amounts (in EUR million) of loans between a bank and a firm in a given month.	36,163,821	-2.318	1.954
Exposure(Feb 2019)	p.p.	Unused exemption allowance, i.e., difference of 6-fold the minimum reserve requirement and the excess liquidity holdings of a bank in February 2019 if such difference is positive, and to zero otherwise. It is expressed in percentage of main assets.	52,814,649	0.648	1.130
Exposure(Oct 2019)	p.p.	Unused exemption allowance, i.e., difference of 6-fold the minimum reserve requirement and the excess liquidity holdings of a bank in October 2019 if such difference is positive, and to zero otherwise. It is expressed in percentage of main assets.	52,814,649	0.520	0.913
Interim(Mar-Oct 2019)	0/1	Dummy variable equal to 1 between March 2019 and October 2019, 0 otherwise.	52,814,649	0.438	0.496
Implementation(Nov 2019-Feb 2020)	0/1	Dummy variable equal to 1 between November 2019 and February 2020, 0 otherwise.	52,814,649	0.228	0.420
Tiering Benefits (Feb 2019)	p.p.	Savings that would have stemmed from holdings of excess liquidity in February 2019 under the tiering, expressed as a ratio of assets, that is, $100 \times [\min(0, DFR \times (RL - MRR \times 6)) - DFR \times RL] \times Assets$.	52,814,649	0.023	0.021
Tiering Benefits (Oct 2019)	p.p.	Savings that would have stemmed from holdings of excess liquidity in October 2019 under the tiering, expressed as a ratio of assets, that is, $100 \times [\min(0, DFR \times (RL - MRR \times 6)) - DFR \times RL] \times Assets$.	52,814,649	0.024	0.019
CDS	p.p.	5-years credit default swaps, in percentage points. One month lag.	52,814,649	1.050	1.182
Excess liquidity	%	Ratio of excess liquidity (current account + deposit facility - minimum reserve requirements) over main assets. One month lag.	52,814,649	4.705	3.754
Holdings of government securities	%	Ratio of holdings of securities issued by general government over main assets. One month lag.	52,814,649	6.613	4.826
Deposit ratio	%	Ratio of deposits from NFCs and households over main liabilities. One month lag.	52,814,649	37.724	21.050
TLTRO funds	%	Ratio of TLTRO uptake over main liabilities. One month lag.	52,814,649	4.212	4.169
Lending rate	% p.a.	Lending rate on outstanding amounts (in % per annum) on loans between a bank and a firm in a given month.	36,163,821	3.129	3.729
Maturity	Days	Residual maturity of loans between a bank and a firm in a given month.	36,163,821	1316	1665
Drawn credit lines	log(EUR mln)	Logarithm of drawn credit lines (in EUR million) between a bank and a firm in a given month.	21,321,876	-3.707	2.674
Undrawn credit lines	log(EUR mln)	Logarithm of granted but undrawn credit lines (in EUR million) between a bank and a firm in a given month.	18,085,424	-4.032	2.546
Overall credit lines	log(EUR mln)	Logarithm of granted (drawn and undrawn) credit lines (in EUR million) between a bank and a firm in a given month.	25,174,025	-3.903	2.362
Panel C. Bank daily panel of the money market transactions					
Variable name	Units	Definition	Obs.	Mean	St. Dev.
Stock of outstanding unsecured borrowing transactions / MRR	Ratio	Stock of outstanding borrowing in the unsecured money market relative to a bank's minimum reserve requirement.	44,269	11.976	16.613
Stock of outstanding secured lending transactions / MRR	Ratio	Stock of outstanding lending in the secured money market relative to a bank's minimum reserve requirement.	44,269	9.776	17.967
Stock of outstanding secured net borrowing transactions / MRR	Ratio	Stock of net borrowing in the secured money market, defined as gross borrowing minus gross lending, relative to a bank's minimum reserve requirement.	44,269	2.200	13.076
CDS spread	p.p.	5-years credit default swaps, in percentage points. Equal to domestic sovereign CDS spread for state-owned banks without issuer-specific CDS.	44,269	1.017	1.719
Interim period (26 Mar 2019 - 29 Oct 2019)	0/1	Dummy variable equal to 1 between 26 March 2019 and 29 October 2019, 0 otherwise.	44,269	0.197	0.398
Implementation (30 Oct 2019 - 28 Jan 2020)	0/1	Dummy variable equal to 1 between 30 October 2019 and 28 January 2019, 0 otherwise.	44,269	0.082	0.275
Exposure in Feb 2019	p.p.	Unused exemption allowance, i.e., difference of 6-fold the minimum reserve requirement and the excess liquidity holdings of a bank between 30 January and 12 March 2019 if such difference is positive, and zero otherwise. It is expressed in percentage of total assets.	44,269	0.493	1.013
Exposure in Oct 2019	p.p.	Unused exemption allowance, i.e., difference of 6-fold the minimum reserve requirement and the excess liquidity holdings of a bank between 18 September and 29 October 2019 if such difference is positive, and zero otherwise. It is expressed in percentage of main total.	44,269	0.426	0.828

(continued on next page)

Table 1 (continued)

Panel C. Bank daily panel of the money market transactions					
Variable name	Units	Definition	Obs.	Mean	St. Dev.
Stock of outstanding unsecured lending transactions / MRR	Ratio	Stock of outstanding lending in the unsecured money market relative to a bank's minimum reserve requirement.	44,269	1.912	4.684
Stock of outstanding unsecured net borrowing transactions / MRR	Ratio	Stock of net borrowing in the unsecured money market, defined as gross borrowing minus gross lending, relative to a bank's minimum reserve requirement.	44,269	7.257	13.375
CDS spread	p.p.	5-years credit default swaps, in percentage points. Equal to domestic sovereign CDS spread for state-owned banks without issuer-specific CDS.	44,269	1.017	1.719
Interim period (26 Mar 2019 - 29 Oct 2019)	0/1	Dummy variable equal to 1 between 26 March 2019 and 29 October 2019, 0 otherwise.	44,269	0.197	0.398
Implementation (30 Oct 2019 - 28 Jan 2020)	0/1	Dummy variable equal to 1 between 30 October 2019 and 28 January 2019, 0 otherwise.	44,269	0.082	0.275
Exposure in Feb 2019	p.p.	Unused exemption allowance, i.e., difference of 6-fold the minimum reserve requirement and the excess liquidity holdings of a bank between 30 January and 12 March 2019 if such difference is positive, and zero otherwise. It is expressed in percentage of total assets.	44,269	0.493	1.013
Exposure in Oct 2019	p.p.	Unused exemption allowance, i.e., difference of 6-fold the minimum reserve requirement and the excess liquidity holdings of a bank between 18 September and 29 October 2019 if such difference is positive, and zero otherwise. It is expressed in percentage of main total.	44,269	0.426	0.828

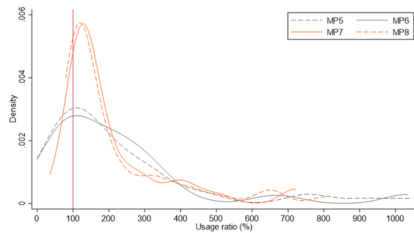


Fig. 2. The distribution of excess liquidity between banks before and after the tiering. The figure plots the distribution of the ratio of a bank's excess liquidity relative to the bank's exemptions equal to six times the MMR in the two maintenance periods before (MP5 and MP6) the tiering implementation in November 2019 and the two after (MP7 and MP8). MP5 goes between July 31, 2019 and September 17, 2019; MP6 between September 18, 2019 and October 29, 2019; MP7 between October 30, 2019 and December 17, 2019; and MP8 between December 18, 2019 and January 28, 2020.

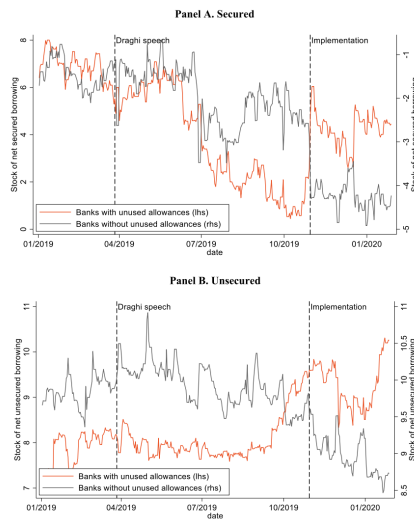


Fig. 3. Net borrowing in the money market. The figure shows the average outstanding stock of net borrowing by banks in EUR billion. The stock of net borrowing is defined as the volume of outstanding borrowing transactions at the end of the day minus the volume of outstanding lending transactions. Panel A is based on transactions in the secured money market segment, and Panel B is based on transactions in the unsecured segment. The data is split between banks with unused tiering allowances (red line, left-hand side axis) and without (grey line, right-hand side axis) during the maintenance period immediately preceding the start of the tiering system at the end of October 2019. Vertical lines mark the speech by President Draghi on March 27, 2019, which first referred to the possibility of introducing a tiering system, as well as to the eventual start of the system on October 30, 2019.

Table 2

Changes in excess liquidity

The table shows results from difference-in-differences regressions of banks' excess liquidity on exposure to the tiering system. The dependent variable in all columns is the bank's monthly change in the ratio of excess liquidity over assets. "Exposure(Feb 2019)" is equal to the unused exemption allowance (as a percentage of main assets) of bank i in February 2019 if such difference is positive, and to zero otherwise. "Exposure(Oct 2019)" is defined in the same way, but as of October 2019. The "Interim (Mar-Oct 2019)" variable is an indicator for the time between the speech by former ECB President Draghi on March 27, 2019, in which he hinted at the introduction of a tiering system for the first time, and the eventual implementation of the system as of October 30, 2019. "Implementation (Nov 2019-Feb 2020)" is an indicator variable for the time after October 30, 2019, i.e., the time since the tiering system has been in place. "CDS" represents banks' CDS spread (in percentage points); for state-owned banks in the sample, this is measured as the domestic sovereign CDS spread. The observation frequency in all regressions is monthly, and the sample period ranges from January 2014 to February 2020. Standard errors are clustered at the bank and country-time level. Standard errors are reported in parentheses. ***, **, and * denote significance at the 1 %, 5 %, and 10 % level, respectively.

Dependent Variable:	(1)	(2)	(3)
Monthly change in excess liquidity			
Exposure(Feb 2019)	-0.059* (0.031)		
Exposure(Feb 2019)*Interim(Mar-Oct 2019)	0.078** (0.030)	0.078** (0.030)	0.078** (0.030)
Exposure(Oct 2019)	0.035 (0.038)		
Exposure(Oct 2019)*Implementation(Nov 2019-Feb 2020)	0.224*** (0.066)	0.224*** (0.066)	0.224*** (0.066)
CDS			0.023 (0.015)
Country-month FE	Yes	Yes	Yes
Bank FE	—	Yes	Yes
Observations	9325	9325	9325
R-squared	0.166	0.178	0.178

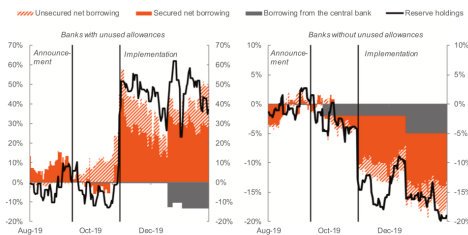


Fig. 4. Change in reserve holdings, money market net borrowing, and central bank borrowing. The figure shows the change in reserve holdings, net borrowing in the money market, and borrowing from the central bank by banks with (left panel) and without (right panel) unused allowances as of October 2019, right before the tiering implementation. The relevant cut-off date for determining the minimum reserve requirements (and therefore the tiering allowance) for the maintenance period ending on October 2019 were July 2019 (for banks reporting monthly) or June 2019 (for banks reporting quarterly). All series are normalized to banks' reserve holdings on September 12, 2019, the date at which the introduction of the two-tier system was formally announced (first vertical dashed line). The second vertical line indicates the start of the tiering system implementation (October 30, 2019). The residual change in reserve holdings can be attributed to other sources of banks' liquidity management, such as the purchase or sale of securities.

- Figure 3 shows a sharp increase in net borrowing by banks with unused allowances around the October 30, 2019 implementation date, especially in the secured market.
- Figure 4 decomposes the reserve reallocation. Banks with unused allowances increased reserve holdings by about 55% from August to November 2019, and roughly 71% of that increase was matched by higher net money-market borrowing. Banks without unused allowances reduced reserve holdings by about 14%, and much of that is matched by additional net lending in money markets.
- Table 3 confirms the pattern econometrically: post-implementation net secured borrowing rises by 1.724 times minimum reserve requirements and net unsecured borrowing by 0.687 times minimum reserve requirements for a one-percentage-point increase in unused allowances.
- Panel C of Table 3 and Figure 6 sharpen the mechanism: banks with unused allowances borrow more from counterparties holding more excess liquidity above their own allowance. Liquidity flows from satiated banks to constrained banks.

Table 3
Money market volumes around the tiering introduction.

Panel A. Bank-level regressions
The table shows results from difference-in-differences regressions of banks' money market activities on the exposure to the tiering system. The dependent variable in all columns is the banks' stock of borrowing, lending, or net borrowing, scaled by their minimum reserve requirements. "Exposure (Feb 2019)" is equal to the maximum of the unused exemption allowance (as a percentage of total assets) in bank and zero between January 30 and March 12, 2019, the last maintenance period before the speech by former ECB President Draghi on March 27, 2019, in which he hinted at the introduction of a tiering system for the first time. "Exposure (Oct 2019)" is defined in the same way, but for the period between September 18 and October 29, 2019, the last maintenance period before the actual implementation of the tiering system. The "Interim (Mar-Oct 2019)" variable is an indicator for the time between the Draghi's speech and the eventual implementation of the system as of October 30, 2019. "Implementation (Nov 2019-Feb 2020)" is an indicator variable for the time between October 30, 2019 and January 28, 2020, i.e., the maintenance periods in which the tiering system was implemented but before the pandemic accelerated in early 2020. "CDS" represents banks' CDS spread (in percentage points) for state-owned banks in the sample, this is measured as the domestic sovereign CDS spread. All regressions include bank fixed effects as well as country-maintenance period fixed effects. The observation frequency is all regressions is daily, and the sample period ranges from January 1, 2017 to January 28, 2020. Robust standard errors (reported in parentheses) are clustered at the bank and maintenance period level. ***, **, and * denote statistical significance at the 1 %, 5 %, and 10 % level, respectively.

	Secured			Unsecured		
	Borrowing (1)	Lending (2)	Net (3)	Borrowing (4)	Lending (5)	Net (6)
Exposure (Feb 2019) x Interim (Mar-Oct 2019)	-0.195 (0.466)	-0.635 (0.394)	0.440 (0.498)	-0.030 (0.207)	-0.039 (0.053)	0.009 (0.202)
Exposure (Oct 2019) x Implementation (Nov 2019-Feb 2020)	0.588 (0.429)	-1.136* (0.583)	1.724** (0.658)	0.551* (0.321)	-0.135 (0.100)	0.687** (0.272)
CDS	-0.766 (0.592)	-0.812 (0.672)	-0.354 (0.996)	1.707 (1.765)	0.067 (0.090)	1.641 (1.696)
Country-MP fixed effects	Y	Y	Y	Y	Y	Y
Bank fixed effects	Y	Y	Y	Y	Y	Y
Observations	44,269	44,269	44,269	44,269	44,269	44,269
No. banks	42	42	42	42	42	42
R ²	0.920	0.910	0.878	0.802	0.939	0.837
R ² (within)	0.002	0.002	0.004	0.006	0.001	0.005

Panel B. Bank-level regressions on money market borrowing contractual features
The table shows results from difference-in-differences regressions of banks' money market activities on the exposure to the tiering system. The dependent variable in columns (1) and (2) is the volume-weighted average interest rate on banks' borrowing in the secured and unsecured money market, respectively, expressed as a spread over the prevailing DFR. In columns (3) and (4), the dependent variable is the volume-weighted average maturity of the outstanding borrowing in the secured and unsecured segment, respectively. All other variables are defined as explained in notes to Panel A. All regressions include bank fixed effects as well as country-maintenance period fixed effects. The observation frequency is all regressions is daily, and the sample period ranges from January 1, 2017 to January 28, 2020. Robust standard errors (reported in parentheses) are clustered at the bank and maintenance period level. ***, **, and * denote statistical significance at the 1 %, 5 %, and 10 % level, respectively.

	Borrowing rate		Borrowing maturity	
	Secured (1)	Unsecured (2)	Secured (3)	Unsecured (4)
Exposure (Feb 2019) x Interim (Mar-Oct 2019)	-0.006 (0.007)	0.014 (0.023)	1.981 (2.670)	-2.787 (3.803)
Exposure (Oct 2019) x Implementation (Nov 2019-Feb 2020)	0.001 (0.012)	0.042 (0.077)	1.885 (1.759)	-4.815 (5.380)
CDS	0.032 (0.020)	0.004 (0.018)	8.685 (9.213)	-0.574 (4.218)
Country-MP fixed effects	Y	Y	Y	Y
Bank fixed effects	Y	Y	Y	Y
Observations	42,527	44,016	42,527	44,016
No. banks	42	42	42	42
R ²	0.696	0.938	0.767	0.714
R ² (within)	0.003	0.010	0.025	0.002

Panel C. Relationship-level regressions
The table shows results from difference-in-differences regressions of banks' unsecured net borrowing on exposure to the tiering system at the bank-counterparty level. The dependent variable in columns (1)-(3) is the banks' stock of outstanding net borrowing per counterparty, and in column (4) the volume-weighted borrowing rate on the outstanding stock of borrowing per counterparty. Variables are defined as explained in notes to Panel A. Column (1) includes bank fixed effects as well as bank's country-maintenance period fixed effects. Column (2) includes bank fixed effects and counterparty-maintenance period fixed effects. Column (3) contains bank-counterparty fixed effects, counterparty-maintenance period fixed effects, and lender's country-maintenance period fixed effects. The observation frequency is all regressions is daily, and the sample period ranges from January 1, 2017 to January 28, 2020. Robust standard errors (reported in parentheses) are clustered at the bank and maintenance period level. ***, **, and * denote statistical significance at the 1 %, 5 %, and 10 % level, respectively.

Dependent variable:	Unsecured net borrowing volumes			Unsecured borrowing rate
	(1)	(2)	(3)	(4)
Exposure (Feb 2019) x Interim (Mar-Oct 2019)	-0.002* (0.001)	0.0199* (0.011)	0.012 (0.009)	-0.003 (0.004)
Exposure (Oct 2019) x Implementation (Nov 2019-Feb 2020)	0.002* (0.001)	0.016* (0.008)	0.009*** (0.003)	-0.005 (0.003)
CDS	0.007 (0.006)	0.009 (0.011)	0.018 (0.010)	0.002 (0.004)
Bank's country-MP fixed effects	Y	Y	Y	Y
Bank fixed effects	Y	Y	Y	Y
Counterparty-MP fixed effects	-	Y	Y	Y
Bank-counterparty fixed effects	-	-	Y	Y
Observations	23,337,146	23,333,780	23,333,780	2,450,325

(continued on next page)

Table 3 (continued)

Panel C. Relationship-level regressions
The table shows results from difference-in-differences regressions of banks' unsecured net borrowing on exposure to the tiering system at the bank-counterparty level. The dependent variable in columns (1)-(3) is the banks' stock of outstanding net borrowing per counterparty, and in column (4) the volume-weighted borrowing rate on the outstanding stock of borrowing per counterparty. Variables are defined as explained in notes to Panel A. Column (1) includes bank fixed effects as well as bank's country-maintenance period fixed effects. Column (2) includes bank fixed effects and counterparty-maintenance period fixed effects. Column (3) contains bank-counterparty fixed effects, counterparty-maintenance period fixed effects, and lender's country-maintenance period fixed effects. The observation frequency is all regressions is daily, and the sample period ranges from January 1, 2017 to January 28, 2020. Robust standard errors (reported in parentheses) are clustered at the bank and maintenance period level. ***, **, and * denote statistical significance at the 1 %, 5 %, and 10 % level, respectively.

Dependent variable:	Unsecured net borrowing volumes			Unsecured borrowing rate
	(1)	(2)	(3)	(4)
No. Banks	42	42	42	42
R ²	0.021	0.231	0.761	0.976
R ² (within)	0.001	0.001	0.001	0.001

5.5 Figure 5 and Table 3, Panel B: No price or maturity rationing in money markets

Read:

- Figure 5 shows that treated banks did not pay meaningfully higher borrowing spreads after they started borrowing more reserves.
- Panel B of Table 3 makes that precise: there is essentially no significant increase in borrowing rates or deterioration in maturities in either secured or unsecured markets.

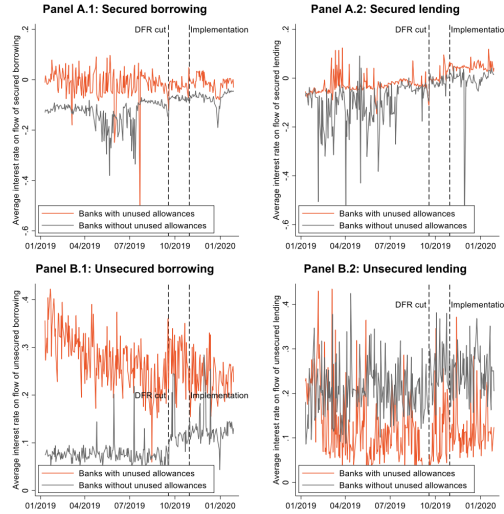


Fig. 5. Money market interest rates
The figure shows the volume-weighted average interest rates on the flow of new money market transactions by reporting banks per day, expressed as a spread over the prevailing DFR. The average is computed across all reporting banks and maturities. The red line indicates the values for banks with unused allowances, and the grey line banks without unused allowances under the tiering system. The vertical lines indicate the date when the reduction of the DFR took effect (September 18, 2019) and the start of the tiering system implementation (October 30, 2019).

- This is a crucial credibility result. If reserve lenders had needed higher compensation, the treatment would have been much harder to interpret as a pure redistribution toward higher-liquidity-need banks.

Table 3
Money market volumes around the tiering introduction.

Panel A. Bank-level regressions
The table shows results from difference-in-differences regressions of banks' money market activities on the exposure to the tiering system. The dependent variable in all columns is the banks' stock of borrowing, lending, or net borrowing, scaled by their minimum reserve requirements. "Exposure (Feb 2019)" is equal to the maximum of the unused exemption allowance (as a percentage of total assets) of bank i and zero between January 20 and March 12, 2019, the last maintenance period before the speech by former ECB President Draghi on March 27, 2019, in which he hinted at the introduction of a tiering system for the first time. "Exposure (Oct 2019)" is defined in the same way, but for the period between September 18 and October 29, 2019, the last maintenance period before the actual implementation of the tiering system. The "Interim (Mar-Oct 2019)" variable is an indicator for the time between the Draghi's speech and the eventual implementation of the system as of October 30, 2019. "Implementation (Nov 2019-Feb 2020)" is an indicator variable for the time between October 30, 2019 and January 28, 2020, i.e., the maintenance periods in which the tiering system was implemented but before the pandemic accelerated in early 2020. "CDS" represents banks' CDS spread (in percentage points) for state-owned banks in the sample, this is measured as the domestic sovereign CDS spread. All regressions include bank fixed effects as well as a country-maintenance period fixed effects. The observation frequency in all regressions is daily, and the sample period ranges from January 1, 2017 to January 28, 2020. Robust standard errors (reported in parentheses) are clustered at the bank and maintenance period level. ***, **, and * denote statistical significance at the 1%, 5%, and 10% level, respectively.

	Secured			Unsecured		
	Borrowing (1)	Lending (2)	Net (3)	Borrowing (4)	Lending (5)	Net (6)
Exposure (Feb 2019) x Interim (Mar-Oct 2019)	-0.195 (0.466)	-0.638 (0.394)	0.440 (0.498)	-0.030 (0.207)	-0.039 (0.053)	0.009 (0.202)
Exposure (Oct 2019) x Implementation (Nov 2019-Feb 2020)	0.588 (0.429)	-1.136* (0.581)	1.724** (0.650)	0.551* (0.321)	-0.135 (0.100)	0.687** (0.272)
CDS	-0.766 (0.592)	-0.412 (0.672)	-0.354 (0.996)	1.707 (1.765)	0.067 (0.090)	1.641 (1.696)
Country-MP fixed effects	Y	Y	Y	Y	Y	Y
Bank fixed effects	Y	Y	Y	Y	Y	Y
Observations	44,269	44,269	44,269	44,269	44,269	44,269
No. Banks	42	42	42	42	42	42
R ²	0.920	0.910	0.878	0.802	0.939	0.837
R ² (within)	0.002	0.002	0.004	0.006	0.001	0.005

Panel B. Bank-level regressions on money market borrowing' contractual features
The table shows results from difference-in-differences regressions of banks' money market activities on the exposure to the tiering system. The dependent variable in column (1) and (2) is the volume-weighted average interest rate on banks' borrowing in the secured and unsecured money market, respectively, expressed as a spread over the prevailing DFR. In columns (3) and (4), the dependent variable is the volume-weighted average maturity of the outstanding borrowing in the secured and unsecured segment, respectively. All other variables are defined as explained in notes to Panel A. All regressions include bank fixed effects as well as country-maintenance period fixed effects. The observation frequency in all regressions is daily, and the sample period ranges from January 1, 2017 to January 28, 2020. Robust standard errors (reported in parentheses) are clustered at the bank and maintenance period level. ***, **, and * denote statistical significance at the 1%, 5%, and 10% level, respectively.

	Borrowing rate		Borrowing maturity	
	Secured (1)	Unsecured (2)	Secured (3)	Unsecured (4)
Exposure (Feb 2019) x Interim (Mar-Oct 2019)	-0.006 (0.007)	0.014 (0.023)	3.981 (2.670)	-2.787 (3.803)
Exposure (Oct 2019) x Implementation (Nov 2019-Feb 2020)	0.001 (0.012)	0.042 (0.027)	1.885 (1.759)	-4.815 (5.380)
CDS	0.032 (0.020)	0.004 (0.018)	6.685 (6.213)	-8.574 (4.218)
Country-MP fixed effects	Y	Y	Y	Y
Bank fixed effects	Y	Y	Y	Y
Observations	42,527	44,016	42,527	44,016
No. Banks	42	42	42	42
R ²	0.696	0.938	0.767	0.714
R ² (within)	0.003	0.010	0.025	0.002

Panel C. Relationship-level regressions
The table shows results from difference-in-differences regressions of banks' unsecured net borrowing on exposure to the tiering system at the bank-counterparty level. The dependent variable in columns (1)-(3) is the banks' stock of outstanding net borrowing per counterparty, and in column (4) the volume-weighted borrowing rate on the outstanding stock of borrowing per counterparty. Variables are defined as explained in notes to Panel A. Column (1) includes bank fixed effects as well as bank's country-maintenance period fixed effects. Column (2) includes bank fixed effects and counterparty-maintenance period fixed effects. Column (3) contains bank-counterparty fixed effects, counterparty-maintenance period fixed effects, and lender's country-maintenance period fixed effects. The observation frequency in all regressions is daily, and the sample period ranges from January 1, 2017 to January 28, 2020. Robust standard errors (reported in parentheses) are clustered at the bank and maintenance period level. ***, **, and * denote statistical significance at the 1%, 5%, and 10% level, respectively.

Dependent variable:	Unsecured net borrowing volumes			Unsecured borrowing rate
	(1)	(2)	(3)	(4)
Exposure (Feb 2019) x Interim (Mar-Oct 2019)	-0.002* (0.001)	0.0199* (0.011)	0.012 (0.009)	-0.003 (0.004)
Exposure (Oct 2019) x Implementation (Nov 2019-Feb 2020)	0.002* (0.001)	0.016* (0.006)	0.009*** (0.003)	-0.005 (0.003)
CDS	0.007 (0.006)	0.009 (0.011)	0.018 (0.016)	0.002 (0.004)
Bank's country-MP fixed effects	Y	Y	Y	Y
Bank fixed effects	Y	Y	Y	Y
Counterparty-MP fixed effects	Y	Y	Y	Y
Bank-counterparty fixed effects	Y	Y	Y	Y
Observations	23,337,146	23,333,780	23,333,780	2,450,325

(continued on next page)

Table 3 (continued)

Panel C. Relationship-level regressions
The table shows results from difference-in-differences regressions of banks' unsecured net borrowing on exposure to the tiering system at the bank-counterparty level. The dependent variable in column (1)-(3) is the banks' stock of outstanding net borrowing per counterparty, and in column (4) the volume-weighted borrowing rate on the outstanding stock of borrowing per counterparty. Variables are defined as explained in notes to Panel A. Column (1) includes bank fixed effects as well as bank's country-maintenance period fixed effects. Column (2) includes bank fixed effects and counterparty-maintenance period fixed effects. Column (3) contains bank-counterparty fixed effects, counterparty-maintenance period fixed effects, and lender's country-maintenance period fixed effects. The observation frequency in all regressions is daily, and the sample period ranges from January 1, 2017 to January 28, 2020. Robust standard errors (reported in parentheses) are clustered at the bank and maintenance period level. ***, **, and * denote statistical significance at the 1%, 5%, and 10% level, respectively.

Dependent variable:	Unsecured net borrowing volumes			Unsecured borrowing rate
	(1)	(2)	(3)	(4)
No. Banks	42	42	42	42
R ²	0.021	0.231	0.761	0.976
R ² (within)	0.001	0.001	0.001	0.001

Table 4

Changes in bank asset composition.

Panel A. Government bond holdings			
The table shows results from difference-in-differences regressions of banks' government bond holdings on exposure to the tiering system. The dependent variable in all columns is the bank's monthly change in the ratio of government bonds over assets. "Exposure(Feb 2019)" is equal to the unused exemption allowance (as a percentage of main assets) of bank i in February 2019 if such difference is positive, and to zero otherwise. "Exposure(Oct 2019)" is defined in the same way, but as of October 2019. The "Interim (Mar-Oct 2019)" variable is an indicator for the time between the speech by former ECB President Draghi on March 27, 2019, in which he hinted at the introduction of a tiering system for the first time, and the eventual implementation of the system as of October 30, 2019. "Implementation (Nov 2019-Feb 2020)" is an indicator variable for the time after October 30, 2019, i.e., the time when the tiering system took effect. "CDS" represents banks' CDS spread (in percentage points); for state-owned banks in the sample, this is measured as the domestic sovereign CDS spread. The observation frequency in all regressions is monthly, and the sample period ranges from January 2014 to February 2020. Standard errors are clustered at the bank and country-time level. Standard errors are reported in parentheses. ***, **, and * denote significance at the 1 %, 5 %, and 10 % level, respectively.			
Dependent Variable:	(1)	(2)	(3)
Monthly change in holdings of government securities			
Exposure(Feb 2019)	0.006 (0.005)		
Exposure(Feb 2019)*Interim(Mar-Oct 2019)	-0.021 (0.012)	-0.020 (0.013)	-0.021 (0.013)
Exposure(Oct 2019)	-0.000 (0.005)		
Exposure(Oct 2019)*Implementation(Nov 2019-Feb 2020)	-0.026** (0.012)	-0.026** (0.012)	-0.026** (0.012)
CDS			-0.016 (0.012)
Country-month FE	Yes	Yes	Yes
Bank FE	-	Yes	Yes
Observations	9,325	9,325	9,325
R-squared	0.208	0.217	0.217

Panel B. Loan volumes

The table shows how bank loan volumes to firms changed after the announcement and implementation of the tiering depending on the banks' exposure to the tiering system. The dependent variable in all columns is the bank's monthly change in the ratio of the stock of outstanding loans to all non-financial corporations over assets. "Exposure(Feb 2019)" is equal to the unused exemption allowance (as a percentage of main assets) of bank i in February 2019 if such difference is positive, and to zero otherwise. "Exposure (Oct 2019)" is defined in the same way, but as of October 2019. The "Interim (Mar-Oct 2019)" variable is an indicator for the time between the speech by former ECB President Draghi on March 27, 2019, in which he hinted at the introduction of a tiering system for the first time, and the eventual implementation of the system as of October 30, 2019. "Implementation (Nov 2019-Feb 2020)" is an indicator variable for the time after October 30, 2019, i.e., the time when the tiering system took effect. "CDS" represents banks' CDS spread (in percentage points); for state-owned banks in the sample, this is measured as the domestic sovereign CDS spread. The observation frequency in all regressions is monthly, and the sample period ranges from January 2014 to February 2020. Standard errors are clustered at the bank and country-time level. Standard errors are reported in parentheses. ***, **, and * denote significance at the 1 %, 5 %, and 10 % level, respectively.

Dependent Variable:	(1)	(2)	(3)
Monthly change in NFC loans			
Exposure(Feb 2019)	0.017* (0.009)		
Exposure(Feb 2019)*Interim(Mar-Oct 2019)	-0.052** (0.022)	-0.052** (0.023)	-0.052** (0.023)
Exposure(Oct 2019)	-0.008 (0.007)		
Exposure(Oct 2019)*Implementation(Nov 2019-Feb 2020)	0.015 (0.047)	0.015 (0.048)	0.015 (0.048)
CDS			0.000 (0.012)
Country-month FE	Yes	Yes	Yes
Bank FE	-	Yes	Yes
Observations	9,325	9,325	9,325
R-squared	0.173	0.182	0.182

5.6 Table 4: Balance-sheet rebalancing before clean loan-supply identification

Read:

- Panel A shows that banks with unused allowances cut government-securities holdings after implementation. The key coefficient is about -0.026 , meaning reserves crowd in lending partly by crowding out liquid securities.
- Panel B shows no clear effect on bank-level outstanding loans in the aggregate balance-sheet data.
- That is *not* evidence against the lending channel. The authors explain that bank-level loan growth is contaminated by loan-demand heterogeneity, which is why the rest of the paper moves to matched bank-firm data.

5.7 Table 5: Main lending-supply results

Read:

- Panel A is the core result. Depending on the fixed-effects specification, the post-implementation coefficient ranges from 0.040 to 0.074. So a one-percentage-point increase in unused allowances raises lending to a given firm by roughly 4% to 7%.
- The result survives the strongest specification with firm-month fixed effects and bank-firm fixed effects, which is the cleanest within-borrower comparison.
- Panel B shows that treated banks form more *new* lending relationships, with coefficients between

0.027 and 0.035, but they do not terminate fewer old relationships. So the lending expansion comes from extra supply rather than from mere relationship preservation.

Table 5
The effects of unused exemptions on bank lending.

Panel A. Outstanding loan volume				
The table shows how banks' lending to firms changes after the announcement and implementation of the tiering depending on the banks' exposure to the tiering system. The dependent variable is the logarithm of loans by bank i to a non-financial corporation f in month t . "ExposureFeb 2019" is equal to the unused exemption allowance (as a percentage of main assets) of bank i in February 2019 if such difference is positive, and to zero otherwise. "ExposureOct 2019" is defined in the same way, but as of October 2019. The "Interim (Mar-Oct 2019)" variable is an indicator for the time between the speech by former ECB President Draghi on March 27, 2019, in which he hinted at the introduction of a tiering system for the first time, and the eventual implementation of the system as of October 30, 2019. "Implementation (Nov 2019-Feb 2020)" is an indicator variable for the time after October 30, 2019, i.e., the time when the tiering system took effect. Control variables are as defined in Table 1. Panel B. The observation frequency in all regressions is monthly, and the sample period ranges from September 2018 to February 2020. Standard errors are clustered at the bank-time-period level. Standard errors are reported in parentheses. ***, **, and * denote significance at the 1 %, 5 %, and 10 % level, respectively.				
Dependent Variable:	(1)	(2)	(3)	(4)
Volume of NFC loans	Log	Log	Log	Log
ExposureFeb 2019*Interim(Mar-Oct 2019)	0.012 (0.011)	0.007 (0.006)	0.013 (0.010)	0.011 (0.009)
ExposureOct 2019*Implementation(Nov 2019-Feb 2020)	0.066*** (0.025)	0.040*** (0.012)	0.074*** (0.017)	0.066*** (0.019)
CDS	-0.049 (0.040)	-0.021 (0.020)	-0.034 (0.022)	-0.045 (0.023)
Excess liquidity	0.010** (0.005)	0.002 (0.002)	0.009** (0.005)	0.006 (0.004)
Holdings of government securities	0.055*** (0.016)	0.026*** (0.009)	0.047*** (0.016)	0.038** (0.016)
Deposit ratio	0.000 (0.000)	-0.000 (0.000)	0.000 (0.000)	0.000 (0.000)
TLTRO funds	0.002 (0.002)	-0.001 (0.001)	0.002 (0.002)	0.001 (0.001)
Bank FE	Yes	Yes	Yes	-
Country-Month FE	Yes	-	-	-
Industry-Location-Size-Month FE	-	Yes	-	-
Firm-Month FE	-	-	Yes	Yes
Bank-Firm FE	-	-	-	Yes
Observations	35,356,355	34,338,371	10,353,666	10,256,326
R-squared	0.084	0.719	0.697	0.935

Panel B. New and terminated lending relationships

The table shows how banks' lending relationships with firms change after the announcement and implementation of the tiering depending on the banks' exposure to the tiering system. In columns (1) to (4), the dependent variable is a dummy that takes the value of 1 if a relation between bank and firm i exists (i.e., the volume of outstanding loans is a strictly positive) but it did not exist in month $t-1$, 0 otherwise. For columns (5) to (8), the dependent variable is a dummy that takes value 1 if a relation between bank and firm i in month t does not exist but it existed in month $t-1$, 0 otherwise. "ExposureFeb 2019" is equal to the unused exemption allowance (as a percentage of main assets) of bank i in February 2019 if such difference is positive, and to zero otherwise. "ExposureOct 2019" is defined in the same way, but as of October 2019. The "Interim (Mar-Oct 2019)" variable is an indicator for the time between the speech by former ECB President Draghi on March 27, 2019, in which he hinted at the introduction of a tiering system for the first time, and the eventual implementation of the system as of October 30, 2019. "Implementation (Nov 2019-Feb 2020)" is an indicator variable for the time after October 30, 2019, i.e., the time when the tiering system took effect. Control variables are as defined in Table 1. Panel B. The observation frequency in all regressions is monthly, and the sample period ranges from September 2018 to February 2020. Standard errors are clustered at the bank-time-period level. Standard errors are reported in parentheses. ***, **, and * denote significance at the 1 %, 5 %, and 10 % level, respectively.

Dependent Variable:	Dummy(New Relationship)				Dummy(Terminated Relationship)			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	0/1	0/1	0/1	0/1	0/1	0/1	0/1	0/1
ExposureFeb 2019*Interim(Mar-Oct 2019)	0.011 (0.008)	0.012 (0.008)	0.016 (0.010)	0.016* (0.010)	-0.000 (0.001)	-0.000 (0.001)	-0.001 (0.001)	-0.001 (0.001)
ExposureOct 2019*Implementation(Nov 2019-Feb 2020)	0.027** (0.012)	0.031** (0.014)	0.035** (0.017)	0.035** (0.017)	0.003 (0.003)	0.001 (0.001)	-0.001 (0.002)	-0.001 (0.002)
CDS	0.052* (0.028)	0.063* (0.034)	0.054* (0.031)	0.054* (0.032)	0.005 (0.005)	0.003 (0.002)	0.001 (0.003)	0.001 (0.003)
Excess liquidity	-0.006 (0.009)	-0.007 (0.009)	-0.007 (0.009)	-0.008 (0.009)	-0.002* (0.001)	-0.000 (0.001)	0.001 (0.001)	0.001 (0.001)
Holdings of government securities	-0.016 (0.010)	-0.016 (0.012)	-0.014 (0.013)	-0.015 (0.014)	-0.001 (0.002)	-0.001* (0.001)	-0.000 (0.001)	-0.000 (0.001)
Deposit ratio	0.001 (0.001)	0.001 (0.001)	0.001 (0.001)	0.000*** (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)
TLTRO funds	-0.000 (0.007)	-0.000 (0.008)	-0.000 (0.010)	-0.000 (0.010)	0.000 (0.001)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)
Bank FE	Yes	Yes	Yes	-	Yes	Yes	Yes	-
Country-Month FE	Yes	-	-	-	Yes	-	-	-
Industry-Location-Size-Month FE	-	Yes	-	-	-	Yes	-	-
Firm-Month FE	-	-	Yes	Yes	-	-	Yes	Yes
Bank-Firm FE	-	-	-	Yes	-	-	-	Yes
Observations	52,814,649	41,330,058	17,903,543	52,814,649	41,330,058	17,903,543	41,330,058	17,903,543
R-squared	0.254	0.362	0.516	0.565	0.034	0.177	0.432	0.481

5.8 Figure 7 and Tables 6 and 7: No pre-trend, not a profitability story, not other ECB policies

Table 6

Alternative channels

The table shows how banks' lending to firms changes after the announcement and implementation of the tiering depending on the banks' exposure to the tiering system. The dependent variable in all columns is the logarithm of loans by bank i to a non-financial corporation f in month t . "ExposureFeb 2019" is equal to the unused exemption allowance (as a percentage of main assets) of bank i in February 2019 if such difference is positive, and to zero otherwise. "ExposureOct 2019" is defined in the same way, but as of October 2019. The "Interim (Mar-Oct 2019)" variable is an indicator for the time between the speech by former ECB President Draghi on March 27, 2019, in which he hinted at the introduction of a tiering system for the first time, and the eventual implementation of the system as of October 30, 2019. "Implementation (Nov 2019-Feb 2020)" is an indicator variable for the time after October 30, 2019, i.e., the time when the tiering system took effect. Tiering benefits in February (October) 2019 are defined as $\text{min}(0, \text{DPR} \times (\text{EL} - \text{MRR} \times 6)) - \text{DPR} \times \text{EL}/\text{Assets}$ in February (October) 2019. Control variables are as defined in Table 1, Panel B. The observation frequency in all regressions is monthly, and the sample period ranges from September 2018 to February 2020. Standard errors are clustered at the bank-time-period level. Standard errors are reported in parentheses. ***, **, and * denote significance at the 1 %, 5 %, and 10 % level, respectively.

Dependent Variable:	(1)	(2)
Volume of NFC loans	Log	Log
ExposureFeb 2019*Interim(Mar-Oct 2019)	0.012 (0.010)	0.015 (0.012)
ExposureOct 2019*Implementation(Nov 2019-Feb 2020)	0.069*** (0.020)	0.075*** (0.023)
Tiering Benefits(Feb 2019)*Interim(Mar-Oct 2019)	0.553 (0.578)	0.553 (0.578)
Tiering Benefits(Oct 2019)*Implementation(Nov 2019-Feb 2020)	0.970 (0.819)	0.970 (0.819)
Excess liquidity(Feb 2019)*Interim(Mar-Oct 2019)	0.003 (0.003)	0.003 (0.003)
Excess liquidity(Oct 2019)*Implementation(Nov 2019-Feb 2020)	0.005 (0.004)	0.005 (0.004)
CDS	-0.043 (0.033)	-0.043 (0.033)
Excess liquidity	0.006 (0.004)	0.006 (0.004)
Holdings of government securities	0.037** (0.016)	0.037** (0.016)
Deposit ratio	0.000 (0.000)	0.000 (0.000)
TLTRO funds	0.003*** (0.001)	0.003*** (0.001)
Bank FE	Yes	Yes
Firm-Month FE	Yes	Yes
Bank-Firm FE	Yes	Yes
Observations	10,256,326	10,256,326
R-squared	0.935	0.935

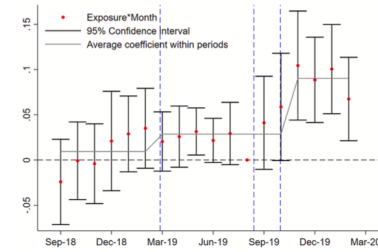


Fig. 7. Dynamic effects of unused exemptions on bank loan supply
In order to test whether our results may be driven by pre-existing trends, we estimate a specification analogous to that in column 3 of Table 7, in which instead of including the terms ExposureFeb 2019*Interim(Mar-Oct 2019) and ExposureOct 2019*Implementation (Nov 2019-Feb 2020), we interact ExposureOct 2019 with time dummies. The figure reports the estimated coefficients on each of these interactions and the 95 % confidence interval.

Table 7

Bank response to monetary policies
The table shows a robustness check on how banks' lending to firms changes after the announcement and implementation of the tiering depending on the banks' exposure to the tiering system, controlling changes in the relation between lending and other control variables. The dependent variable is the logarithm of loans by bank i to a non-financial corporation f in month t . "ExposureFeb 2019" is equal to the unused exemption allowance (as a percentage of main assets) of bank i in February 2019 if such difference is positive, and to zero otherwise. "ExposureOct 2019" is defined in the same way, but as of October 2019. The "Interim (Mar-Oct 2019)" variable is an indicator for the time between the speech by former ECB President Draghi on March 27, 2019, in which he hinted at the introduction of a tiering system for the first time, and the eventual implementation of the system as of October 30, 2019. "Implementation (Nov 2019-Feb 2020)" is an indicator variable for the time after October 30, 2019, i.e., the time when the tiering system took effect. Control variables are as defined in Table 1. Panel B. The observation frequency in all regressions is monthly, and the sample period ranges from September 2018 to February 2020. Standard errors are clustered at the bank-time-period level. Standard errors are reported in parentheses. ***, **, and * denote significance at the 1 %, 5 %, and 10 % level, respectively.

Dependent Variable:	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Volume of NFC loans	Log	Log	Log	Log	Log	Log	Log	Log
ExposureFeb 2019*Interim(Mar-Oct 2019)	0.014 (0.011)	0.014 (0.009)	0.014 (0.009)	0.014 (0.009)	0.014 (0.009)	0.014 (0.009)	0.014 (0.009)	0.014 (0.009)
ExposureOct 2019*Implementation(Nov 2019-Feb 2020)	0.067 (0.021)	0.067*** (0.021)	0.067*** (0.021)	0.067*** (0.021)	0.067*** (0.021)	0.067*** (0.021)	0.067*** (0.021)	0.067*** (0.021)
CDS	-0.051 (0.040)	-0.051 (0.040)	-0.051 (0.040)	-0.051 (0.040)	-0.051 (0.040)	-0.051 (0.040)	-0.051 (0.040)	-0.051 (0.040)
Excess liquidity	0.011 (0.005)	0.011 (0.005)	0.011 (0.005)	0.011 (0.005)	0.011 (0.005)	0.011 (0.005)	0.011 (0.005)	0.011 (0.005)
Holdings of government securities	0.027*** (0.007)	0.027*** (0.007)	0.027*** (0.007)	0.027*** (0.007)	0.027*** (0.007)	0.027*** (0.007)	0.027*** (0.007)	0.027*** (0.007)
Deposit ratio	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)
TLTRO funds	0.002 (0.002)	0.002 (0.002)	0.002 (0.002)	0.002 (0.002)	0.002 (0.002)	0.002 (0.002)	0.002 (0.002)	0.002 (0.002)
Bank FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country-Month FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry-Location-Size-Month FE	-	-	-	-	-	-	-	-
Firm-Month FE	-	-	-	-	-	-	-	-
Bank-Firm FE	-	-	-	-	-	-	-	-
Observations	35,356,355	34,338,371	10,353,666	35,356,355	34,338,371	10,353,666	35,356,355	34,338,371
R-squared	0.084	0.719	0.697	0.084	0.719	0.697	0.084	0.719

Read:

- Figure 7 shows the dynamic event-time coefficients. The lending effect appears only after the reform is implemented; there is no comparable pre-trend.
- Table 6 shows that the interaction of lending with TieringBenefits is insignificant. That weakens the pure profit-relief interpretation.

- Table 6 also shows that banks with high ex ante reserves do not increase lending after the policy, which supports the reserve-distribution story over a simple accounting-income story.
- Table 7 adds interactions with excess liquidity, government securities, deposit ratios, TLTRO use, size, and capital. The main post coefficient remains between about 0.045 and 0.081. So the treatment is not just picking up concurrent ECB actions.

5.9 Tables 8 and 9: The mechanism runs through funding-constrained banks

Table 8

Changes in lending and banks' ex ante money market borrowing rates
The table shows results from difference-in-differences regressions of banks' lending to firms on exposure to the tiering system. In columns (1) and (2), banks are split depending on whether their borrowing rate in the secured money market in October 2019 was above or below the median. In column (3), we test for differences in lending behavior for banks with borrowing rates above and below the median in a pooled sample. "Exposure(Feb 2019)" is equal to the unused exemption allowance (as a percentage of main assets) of bank i in February 2019 if such difference is positive, and to zero otherwise. "Exposure (Oct 2019)" is defined in the same way, but as of October 2019. The "Interim (Mar-Oct 2019)" variable is an indicator for the time between the speech by former ECB President Draghi on March 27, 2019, in which he hinted at the introduction of a tiering system for the first time, and the eventual implementation of the system as of October 30, 2019. "Implementation (Nov 2019-Feb 2020)" is an indicator variable for the time after October 30, 2019, i.e., the time when the tiering system took effect. Control variables include CDS, excess liquidity, holdings of government securities, deposit ratio and TLTRO funds, and are as defined in Table 1, Panel B. The observation frequency in all regressions is monthly, and the sample period ranges from September 2018 to February 2020. Standard errors are clustered at the bank-time-period level. ***, **, and * denote significance at the 1 %, 5 %, and 10 % level, respectively.

Dependent Variable:	(1)	(2)	(3)
	Subsample Banks with borrowing rates		
Volume of NFC loans	Above median	Below median	Pooled
Above median money market rate (Oct-2019):			
Exposure(Feb 2019)*Interim(Mar-Oct 2019)	0.010***		0.008**
Exposure(Oct 2019)	(0.002)		(0.003)
*Implementation(Nov 2019-Feb 2020)	0.152***		0.148***
	(0.016)		(0.009)
Below median money market rate (Oct-2019):			
Exposure(Feb 2019)*Interim(Mar-Oct 2019)		0.056*	0.044*
Exposure(Oct 2019)		(0.031)	(0.025)
*Implementation(Nov 2019-Feb 2020)		-0.005	-0.015
		(0.028)	(0.026)
Controls	Yes	Yes	Yes
Firm-Month FE	Yes	Yes	Yes
Bank-Firm FE	Yes	Yes	Yes
Observations	1,453,670	232,868	2,001,748
R-squared	0.938	0.957	0.942

Table 9
Bank and borrower cross-sectional differences
The table shows results from difference-in-differences regressions of banks' lending to firms on the banks' exposure to the tiering system. Each column reports two separate regressions. We report estimates for the subsample above and below the median of the characteristics indicated in each column. The third panel in each column reports the value of $\hat{\rho}$ and $\hat{\sigma}$ for the significance of the difference (testing significance is indicated by the asterisk between the coefficients in the regressions reported above. "Exposure(Feb 2019)" is equal to the unused exemption allowance (as a percentage of main assets) of bank i in February 2019 if such difference is positive, and to zero otherwise. "Exposure(Oct 2019)" is defined in the same way, but as of October 2019. The "Interim (Mar-Oct 2019)" variable is an indicator for the time between the speech by former ECB President Draghi on March 27, 2019, in which he hinted at the introduction of a tiering system for the first time, and the eventual implementation of the system as of October 30, 2019. "Implementation (Nov 2019-Feb 2020)" is an indicator variable for the time after October 30, 2019, i.e., the time when the tiering system took effect. Control variables include CDS, excess liquidity, holdings of government securities, deposit ratio and TLTRO funds, and are as defined in Table 1, Panel B. The observation frequency in all regressions is monthly, and the sample period ranges from September 2018 to February 2020. Standard errors are clustered at the bank-time-period level. Standard errors are reported in parentheses. ***, **, and * denote significance at the 1 %, 5 %, and 10 % level, respectively.

Dependent Variable: Volume of NFC loans	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Bank capital	Bank CDS	Firm size	Firm age	Firm ROA	Firm leverage	Firm productivity
Sample split by:	Bank capital	Bank CDS	Firm size	Firm age	Firm ROA	Firm leverage	Firm productivity
High							
Exposure(Feb 2019)*Interim(Mar-Oct 2019)	-0.001	0.013	0.001	0.009	0.008	0.009	0.013
Exposure(Oct 2019)*Implementation(Nov 2019-Feb 2020)	0.000	0.012	0.013	0.006	0.001	0.001	0.006
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Low							
Exposure(Feb 2019)*Interim(Mar-Oct 2019)	0.009	0.002	0.017	0.003	0.006	0.006	0.009
Exposure(Oct 2019)*Implementation(Nov 2019-Feb 2020)	0.007	0.011	0.012	0.006	0.001	0.001	0.006
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
F-test: High - Low	3.10*	0.71	2.34	2.62	6.54	1.45	5.51**
Exposure(Feb 2019)*Interim(Mar-Oct 2019)	1.42**	1.24**	0.88	0.88	0.79	12.26***	0.84
Exposure(Oct 2019)*Implementation(Nov 2019-Feb 2020)	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm-Month FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Bank-Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Read:

- Table 8 splits banks by pre-implementation money-market borrowing rates. The lending effect is very large for banks above the median borrowing rate: 0.152 in the high-rate subsample and 0.148 in the pooled interaction specification. It is essentially zero for banks below the median.
- Table 9 shows the same pattern with other proxies for funding costs. Low-capital banks and high-CDS banks drive the effect.
- On the borrower side, the extra lending is not concentrated in obviously riskier or less productive firms. The one mild tilt is toward more levered firms, but there is no broad pattern of lending to low-ROA, low-productivity, or high-default-probability borrowers.
- This is exactly the paper's mechanism: the reform relaxed a liquidity constraint for certain banks rather than triggering indiscriminate risk-taking.

5.10 Tables 10, 11, and 12: Loan prices, maturities, and credit lines

Read:

- Table 10 shows no strong average effect on lending rates across all banks, but banks with high ex ante funding costs cut loan rates. The post coefficient for above-median-rate banks is about -0.048 , roughly a 4.8-basis-point decline per one-percentage-point increase in unused allowances.
- Table 11 shows a substantial increase in maturity: the overall post coefficient is about 24.8 days, and the effect is concentrated among funding-constrained banks.
- Table 12 shows larger drawn, undrawn, and total credit lines. Since credit lines create uncertain future liquidity needs for banks, this is strong evidence that extra reserves reduced precautionary behavior.
- Taken together, the treated banks do not just lend more quantity. They also offer better terms on dimensions that are especially sensitive to liquidity risk.

Table 10

Changes in lending rates

The table shows results from difference-in-differences regressions of the lending rates on banks' exposure to the tiering system. The dependent variable is the average lending rate for loans from bank i to non-financial corporation j in month t . In columns (2) to (4), banks are split depending on whether their borrowing rate in the money market in October 2019 was above or below the median across banks. "Exposure(Feb 2019)" is equal to the unused exemption allowance (as a percentage of main assets) of bank i in February 2019 if such difference is positive, and to zero otherwise. "Exposure(Oct 2019)" is defined in the same way, but as of October 2019. The "Interim (Mar-Oct 2019)" variable is an indicator for the time between the speech by former ECB President Draghi on March 27, 2019, in which he hinted at the introduction of a tiering system for the first time, and the eventual implementation of the system as of October 30, 2019. "Implementation (Nov 2019-Feb 2020)" is an indicator variable for the time after October 30, 2019, i.e., the time when the tiering system took effect. Control variables include CDS, excess liquidity, holdings of government securities, deposit ratio and TLTRO funds, and are as defined in Table 1, Panel B. The observation frequency in all regressions is monthly, and the sample period ranges from September 2018 to February 2020. Standard errors are clustered at the bank-time-period level. Standard errors are reported in parentheses. ***, **, and * denote significance at the 1 %, 5 %, and 10 % level, respectively.

Dependent Variable:	(1)	(2)	(3)	(4)
Interest rate on NFC loans	Overall	Above median	Below median	Pooled
Exposure(Feb 2019)*Interim(Mar-Oct 2019)	-0.017 (0.031)			
Exposure(Oct 2019)*Implementation(Nov 2019-Feb 2020)	0.041 (0.066)			
Above median money market rate (Oct 2019):				
Exposure(Feb 2019)*Interim(Mar-Oct 2019)		-0.013*** (0.003)		-0.018** (0.006)
Exposure(Oct 2019)*Implementation(Nov 2019-Feb 2020)		-0.048*** (0.011)		-0.046*** (0.012)
Below median money market rate (Oct 2019):				
Exposure(Feb 2019)*Interim(Mar-Oct 2019)			0.195 (0.235)	0.156 (0.159)
Exposure(Oct 2019)*Implementation(Nov 2019-Feb 2020)			0.372 (0.401)	0.312 (0.315)
Controls	Yes	Yes	Yes	Yes
Firm-Month FE	Yes	Yes	Yes	Yes
Bank-Firm FE	Yes	Yes	Yes	Yes
Observations	10,256,326	1,453,670	232,868	2,001,748
	0.889	0.907	0.918	0.913

Table 11

Changes in loan maturity

The table shows results from difference-in-differences regressions of the loan maturity on banks' exposure to the tiering system. The dependent variable is the average residual maturity expressed in days for loans from bank i to non-financial corporation j in month t . In columns (2) and (3), banks are split depending on whether their borrowing rate in the money market in October 2019 was above or below the median across banks. "Exposure(Feb 2019)" is equal to the unused exemption allowance (as a percentage of main assets) of bank i in February 2019 if such difference is positive, and to zero otherwise. "Exposure(Oct 2019)" is defined in the same way, but as of October 2019. The "Interim (Mar-Oct 2019)" variable is an indicator for the time between the speech by former ECB President Draghi on March 27, 2019, in which he hinted at the introduction of a tiering system for the first time, and the eventual implementation of the system as of October 30, 2019. "Implementation (Nov 2019-Feb 2020)" is an indicator variable for the time after October 30, 2019, i.e., the time when the tiering system took effect. Control variables include CDS, excess liquidity, holdings of government securities, deposit ratio and TLTRO funds, and are as defined in Table 1, Panel B. The observation frequency in all regressions is monthly, and the sample period ranges from September 2018 to February 2020. Standard errors are clustered at the bank-time-period level. Standard errors are reported in parentheses. ***, **, and * denote significance at the 1 %, 5 %, and 10 % level, respectively.

Dependent Variable:	(1)	(2)	(3)	(4)
Maturity	Overall	Above median	Below median	Pooled
Exposure(Feb 2019)*Interim(Mar-Oct 2019)	3.029 (3.925)			
Exposure(Oct 2019)*Implementation(Nov 2019-Feb 2020)	24.769*** (6.306)			
Above median money market rate (Oct 2019):				
Exposure(Feb 2019)*Interim(Mar-Oct 2019)		-4.593** (1.804)		-4.142*** (1.541)
Exposure(Oct 2019)*Implementation(Nov 2019-Feb 2020)		23.562*** (1.820)		26.843*** (2.196)
Below median money market rate (Oct 2019):				
Exposure(Feb 2019)*Interim(Mar-Oct 2019)			-22.346 (18.546)	-18.934 (11.554)
Exposure(Oct 2019)*Implementation(Nov 2019-Feb 2020)			-30.895* (17.645)	-34.743** (13.077)
Controls	Yes	Yes	Yes	Yes
Firm-Month FE	Yes	Yes	Yes	Yes
Bank-Firm FE	Yes	Yes	Yes	Yes
Observations	10,256,326	1,453,670	232,868	2,001,748
R-squared	0.966	0.907	0.918	0.915

5.11 Table 13: Firm-level borrowing and aggregate effects

Read:

- Firms more exposed to treated banks increase total borrowing at the firm level. The coefficients, around 0.043 to 0.056 in the full sample, are similar in magnitude to the bank-firm estimates in Table 5.
- The effect is particularly strong for firms borrowing from banks with above-median pre-implementation money-market rates.
- This table is what allows the authors to go from "treated banks lend more" to "aggregate credit supply rises." Using these estimates and the size of reserve reallocation, they conclude that one euro of reserves reallocated toward low-reserve banks raises lending by about 15 cents.

6 Short summary

- **Conclusion.** The paper shows that the distribution of central bank reserves matters for credit supply. Moving reserves toward banks that were ex ante short of liquidity expands lending.

Table 12

Credit lines

The table shows results from difference-in-differences regressions of banks' credit lines (drawn in columns (1) and (4), undrawn in columns (2) and (5), and overall in columns (3) and (6)) to firms on the banks' exposure to the tiering system. In columns (1) to (6), we distinguish between banks depending on whether their borrowing rate in the money market in October 2019 was above or below the median across banks. "ExposureFeb 2019" is equal to the unused exemption allowance (as a percentage of main assets) of bank i in February 2019 if such difference is positive, and to zero otherwise. "ExposureOct 2019" is defined in the same way, but as of October 2019. The "interim(Mar-Oct 2019)" variable is an indicator for the time between the speech by former ECB President Draghi on March 27, 2019, in which he hinted at the introduction of a tiering system for the first time, and the eventual implementation of the system as of October 30, 2019. "Implementation(Nov 2019-Feb 2020)" is an indicator variable for the time after October 30, 2019, i.e., the time when the tiering system took effect. Control variables include CDS, excess liquidity, holdings of government securities, deposit ratio and TLTRO funds, and are as defined in Table 1, Panel B. The observation frequency in all regressions is monthly, and the sample period ranges from September 2018 to February 2020. Standard errors are clustered at the bank-time-period level. Standard errors are reported in parentheses. ***, **, and * denote significance at the 1 %, 5 %, and 10 % level, respectively.

Dependent Variable:	(1) Drawn credit lines	(2) Undrawn credit lines	(3) Overall credit lines	(4) Drawn credit lines	(5) Undrawn credit lines	(6) Overall credit lines
ExposureFeb 2019*interim(Mar-Oct 2019)	-0.003 (0.007)	-0.009 (0.008)	-0.000 (0.006)			
ExposureOct 2019*implementation(Nov 2019-Feb 2020)	0.025** (0.010)	0.030* (0.012)	0.031*** (0.010)			
Above median money market rate (Oct 2019): ExposureFeb 2019*interim(Mar-Oct 2019)				0.006* (0.003)	-0.004 (0.006)	0.002 (0.005)
ExposureOct 2019*implementation(Nov 2019-Feb 2020)				0.018** (0.007)	0.020** (0.008)	0.022*** (0.006)
Below median money market rate (Oct 2019): ExposureFeb 2019*interim(Mar-Oct 2019)				-0.016 (0.053)	0.099 (0.119)	-0.013 (0.036)
ExposureOct 2019*implementation(Nov 2019-Feb 2020)				-0.041 (0.156)	0.046 (0.125)	0.005 (0.021)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Firm-Month FE	Yes	Yes	Yes	Yes	Yes	Yes
Bank-Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	4,936,816	3,910,966	5,779,814	929,340	724,860	1,143,426
R-squared	0.934	0.913	0.954	0.944	0.937	0.965

Table 13

Firm level borrowing

The table shows how loan volumes borrowed by firms changed after the announcement and implementation of the tiering depending on the firms' exposure to banks exposed to the tiering system. The dependent variable is the logarithm of loan volume borrowed by firm f in month t from all banks. All dependent variables are obtained by averaging the corresponding firm-bank-month level variables at the firm-month level, with weights equal to the exposures of each firm to each bank the month before, and are labeled with the prefix avg. In columns (1), (2) and (3), all firms are considered in the regression. In Columns (4) and (5), firms are split depending on whether the average borrowing rate in the secured money market in October 2019 of the banks each firm had exposures with the month before was above or below the median. "ExposureFeb 2019" is equal to the unused exemption allowance (as a percentage of main assets) of bank i in February 2019 if such difference is positive, and to zero otherwise. "ExposureOct 2019" is defined in the same way, but as of October 2019. The "interim(Mar-Oct 2019)" variable is an indicator for the time between the speech by former ECB President Draghi on March 27, 2019, in which he hinted at the introduction of a tiering system for the first time, and the eventual implementation of the system as of October 30, 2019. "Implementation(Nov 2019-Feb 2020)" is an indicator variable for the time after October 30, 2019, i.e., the time when the tiering system took effect. Control variables include CDS, excess liquidity, holdings of government securities, deposit ratio and TLTRO funds and are as defined in Table 1, Panel B. Observations are at the firm level. The observation frequency in all regressions is monthly, and the sample period ranges from September 2018 to February 2020. Standard errors are clustered at the main bank-time-period level, where the main bank for each firm is the one with the largest outstanding loan. Standard errors are reported in parentheses. ***, **, and * denote significance at the 1 %, 5 %, and 10 % level, respectively.

Dependent Variable: Volume of NFC loans	(1) Full sample	(2) Full sample	(3) Full sample	(4) Banks with above median money market rate	(5) Banks with below median money market rate
Avg ExposureFeb 2019*interim(Mar-Oct 2019)	0.007 (0.010)	0.020 (0.017)	0.017 (0.020)	0.011 (0.014)	-0.003 (0.016)
Avg ExposureOct 2019*implementation(Nov 2019-Feb 2020)	0.043** (0.019)	0.056*** (0.019)	0.055*** (0.039)	0.141*** (0.016)	0.031* (0.016)
Avg CDS spread	-0.003 (0.013)	0.044*** (0.016)	0.046*** (0.016)	-0.060 (0.048)	0.041 (0.038)
Avg Excess liquidity	0.003 (0.002)	0.008* (0.005)	0.010** (0.004)	0.017*** (0.007)	-0.005 (0.004)
Avg Holdings of government securities	0.021*** (0.000)	0.022*** (0.006)	0.023*** (0.005)	0.026*** (0.010)	0.014 (0.010)
Avg Deposit ratio	0.000*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.002*** (0.001)	0.000 (0.000)
Avg TLTRO funds	0.001 (0.001)	0.002* (0.001)	0.003** (0.001)	-0.004 (0.002)	0.000 (0.002)
Firm FE	Yes	Yes	Yes	Yes	Yes
Country-Month FE	Yes	Yes	Yes	Yes	Yes
Main Bank-Month FE, Sector-Month FE, Location-Month FE	Yes	Yes	Yes	Yes	Yes
Main Bank-Industry-Location FE	26,206,015 0.910	26,205,638 0.915	25,237,538 0.923	8,357,258 0.903	6,865,878 0.903

- **Intuition.** Banks with high funding costs normally hold too few reserves and therefore behave cautiously. Tiering raises the return on reserves for exactly those banks, so they borrow reserves, sell some liquid securities, and lend more.
- **Empirical lesson.** To identify reserve effects you need both a source of exogenous cross-bank variation in reserve demand and borrower-level data rich enough to net out loan demand.

7 Conclusion

7.1 Core conclusion

- The ECB's tiering reform changed the *marginal value* of reserves for banks with unused allowances and thereby reallocated reserves within the banking system.
- That reallocation happened mainly through the money market and did not trigger noticeable price or maturity rationing.
- Banks that received reserves reduced government-bond holdings and expanded credit supply to firms.
- The lending effect is concentrated among banks that looked liquidity-constrained or funding-constrained before the reform.
- There is little evidence that the effect is driven by simple profitability relief, by other simultaneous ECB policies, or by a misallocation of credit toward obviously worse firms.
- Quantitatively, the reallocation is meaningful: one euro of reserves shifted toward low-reserve banks raises lending by roughly 15 cents.

7.2 Economic intuition

- Reserves are the safest and most liquid bank asset. They insure banks against cash shortfalls in a way that even government bonds cannot fully replicate.
- When a bank has high funding costs, it may rationally choose to hold too few reserves in normal times. That leaves it underinsured against liquidity shocks and less willing to lend.

- Tiering fixes that margin for the affected banks: by making reserves more attractive, it lowers precautionary hoarding and lets those banks transform balance-sheet capacity into loans.
- Banks already swimming in reserves are near satiation, so taking reserves away from them has small lending consequences relative to giving reserves to constrained banks.

7.3 Takeaway

This is a paper on an important monetary-transmission question. The big message is not simply that “more reserves raise lending.” It is that the *location* of reserves in the banking system matters. In a world of quantitative tightening, that is exactly the distinction central banks need to care about.